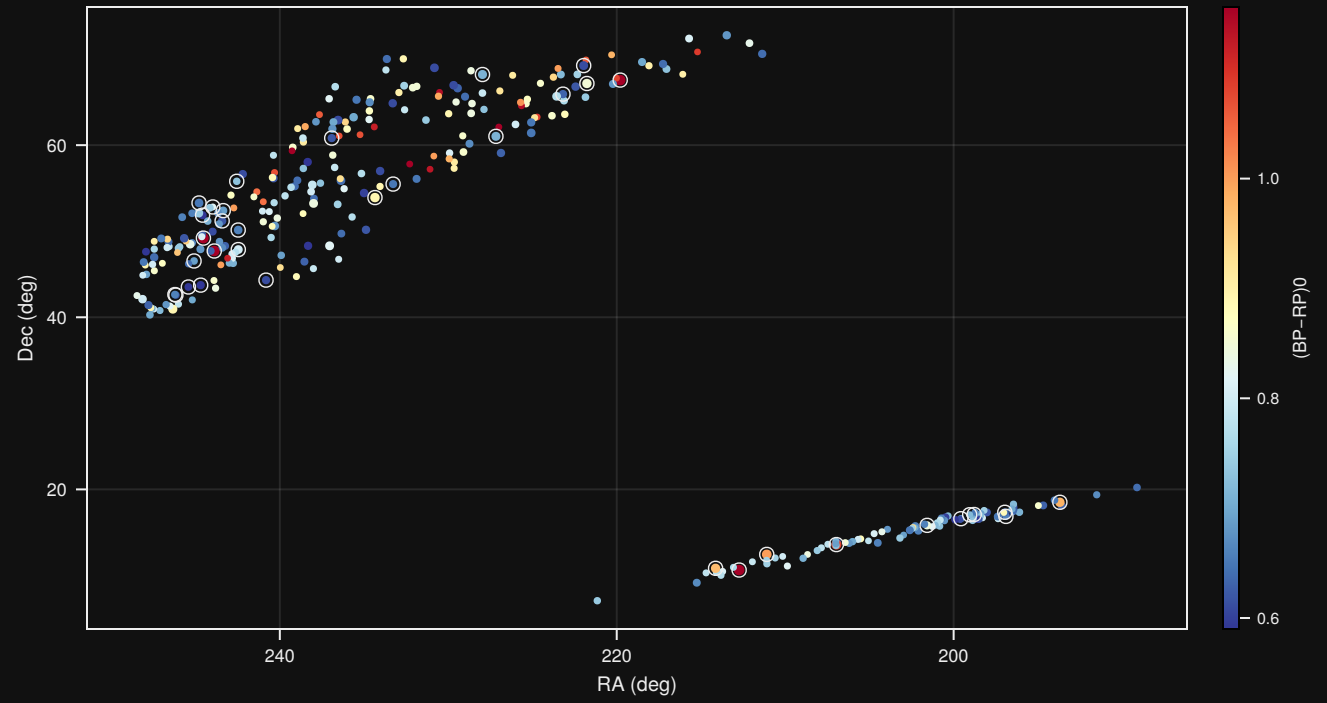
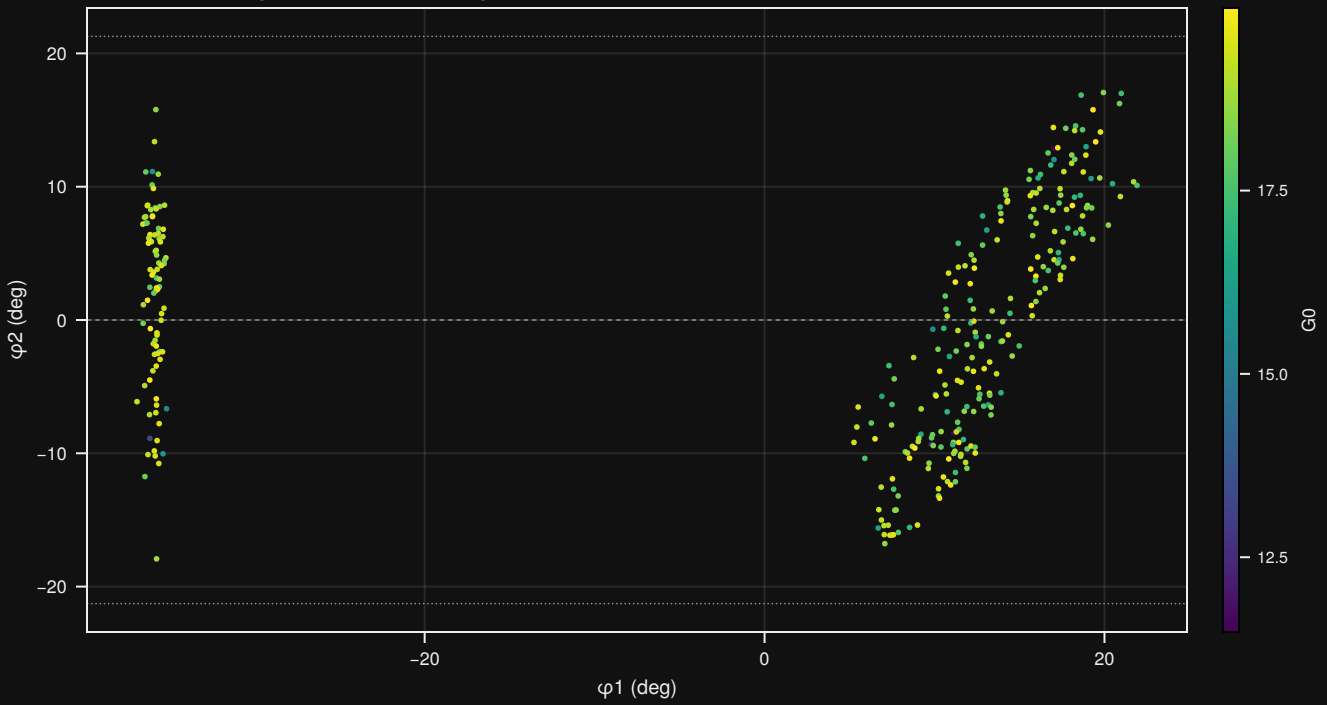
M5

MEMBERSHIP	
Stars in catalog	324
with Gaia GSP-Phot	169 (52%)
with radial velocity	36
Stream id (sID)	62, 65
GEOMETRY · great-circle fit	
Centroid RA, Dec	227.0, 49.7 deg
Centroid l, b	65.4, 46.5 deg
Pole RA, Dec	108.4, 22.4 deg
Length (φ1, 2–98%)	56.5 deg
Width σ(φ2), robust	10.64 deg
KINEMATICS	
pmRA* med ± σ	-8.98 ± 6.81 mas/yr
pmDE med ± σ	4.45 ± 2.75 mas/yr
μφ1* med ± σ	-3.82 ± 1.48 mas/yr
μφ2 med ± σ	9.51 ± 8.09 mas/yr
Parallax (cat.) med	0.178 ± 0.153 mas
Parallax (wtd, ZP-corr.)	0.200 ± 0.011 mas
Distance	d = 5.0 ± 0.3 kpc
Physical length	4.93 kpc
RV median ± σ(MAD)	-361.7 ± 58.5 km/s
RV gradient dV/dφ1	-6.69 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	11.5–20.0 (18.9)
(BP−RP)0 median	0.76
GSP-Phot [M/H] (n=169)	-1.71 ± 0.64
GSP-Phot Teff median	6055 K
RUWE > 1.4	0.0 %
PM clusters (largest)	2 (233 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	324 | 91
n_RV catalog | paper	36 | 11
<[Fe/H]> paper	-1.29
Stellar mass (paper)	710 Msun
Pole offset vs paper	89.8 deg
LABEL CHECK	n differs from paper!

Sky map · catalog members (n = 324) · size ∝ brightness · ring = has RV

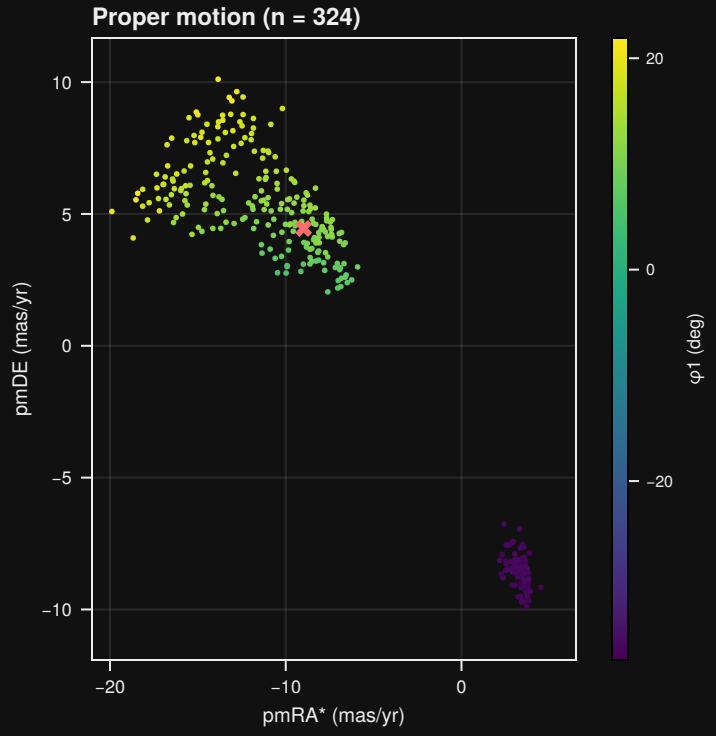


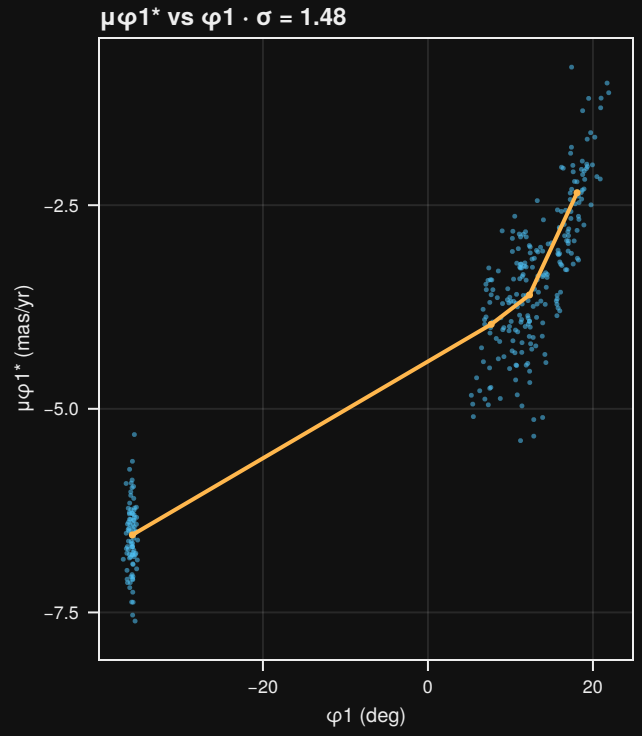
Stream frame (great-circle fit) · length 56.5° · width σ 10.64°

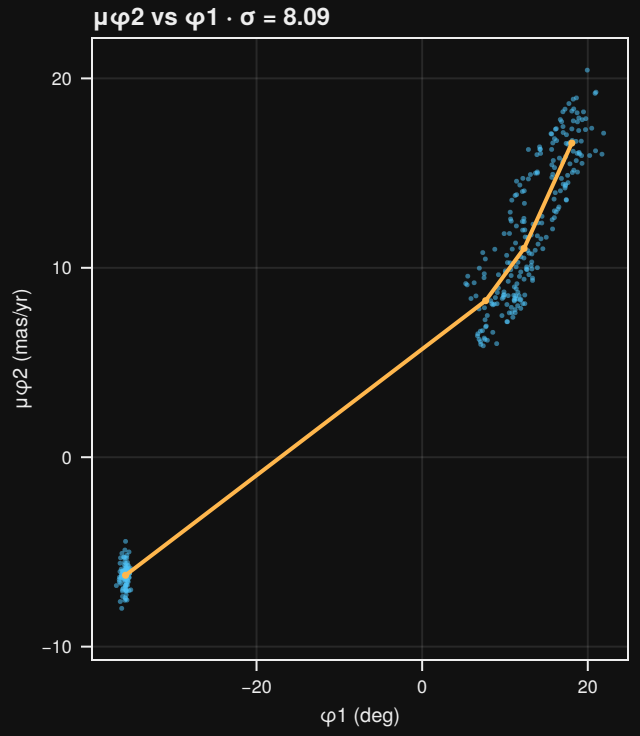


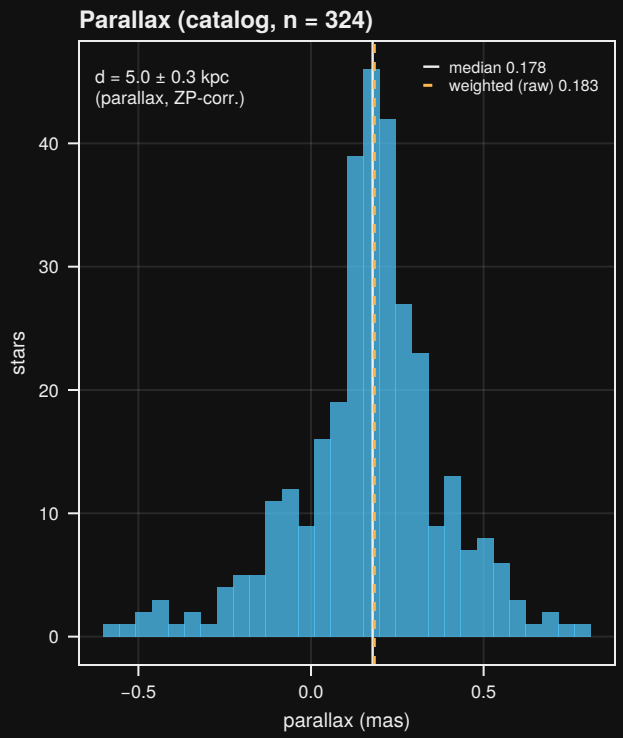
M5

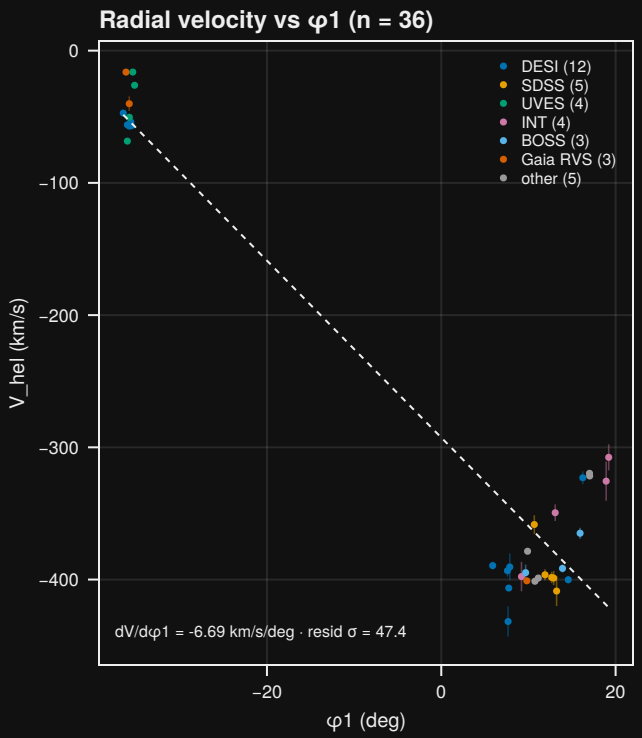
Kinematics · proper motion · parallax · radial velocity

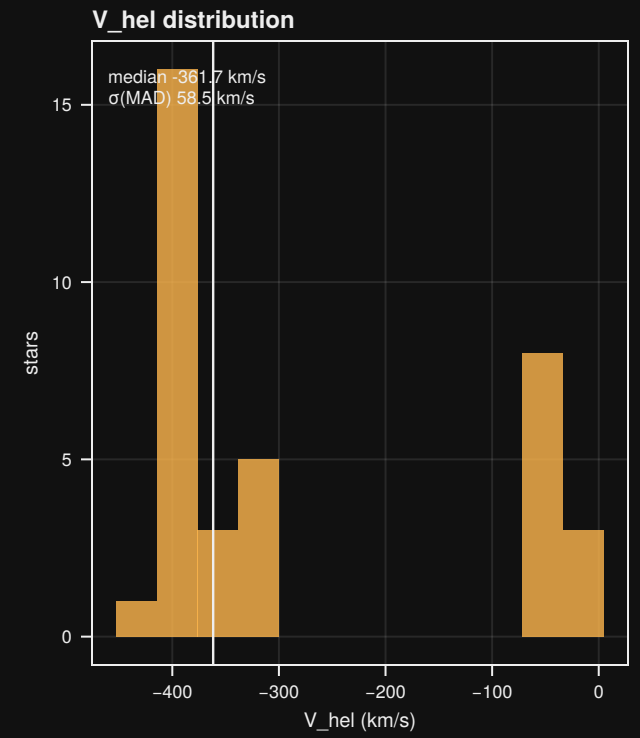






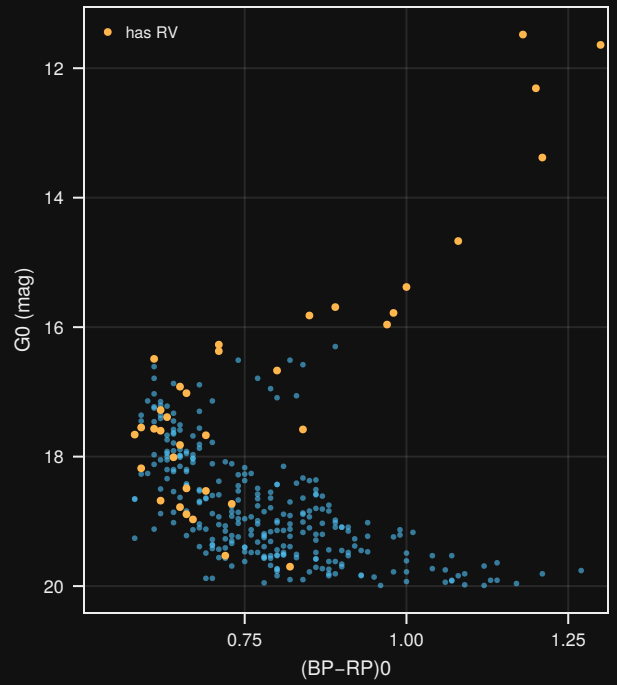




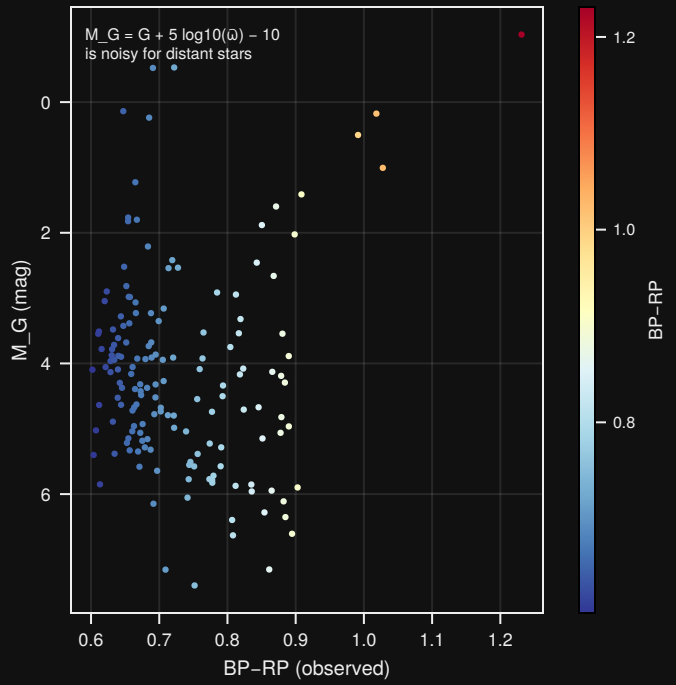


M5

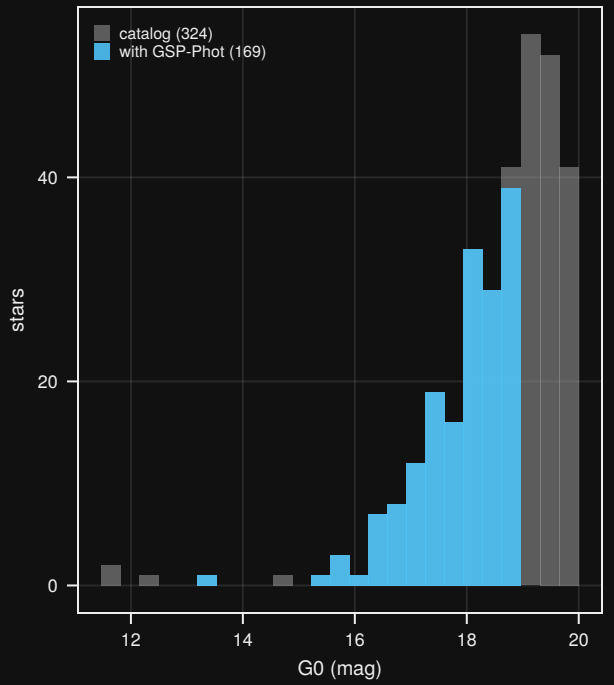
CMD · dereddened (n = 324)



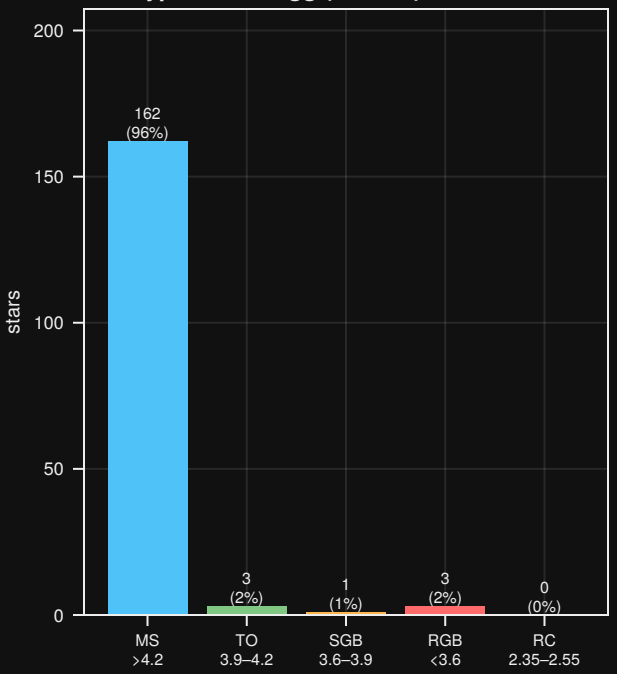
HR diagram · ϖ>0 (n = 158/169)



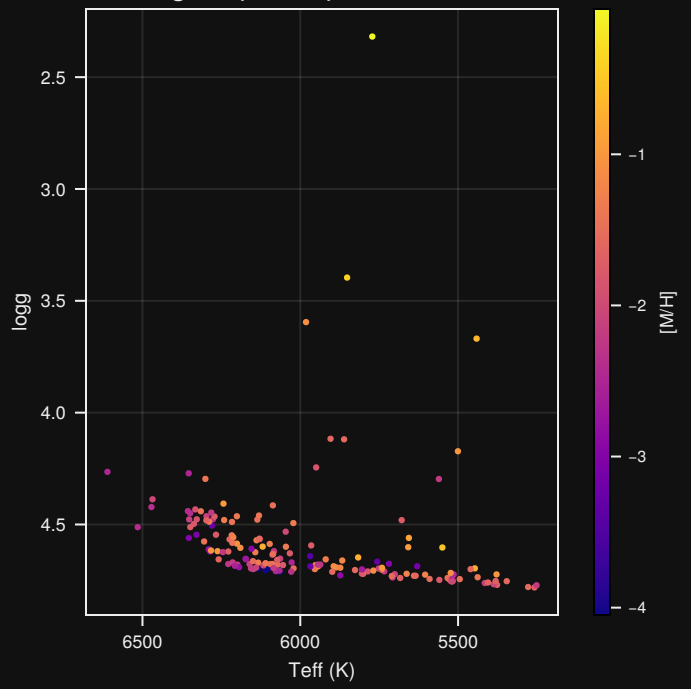
G0 & GSP-Phot coverage



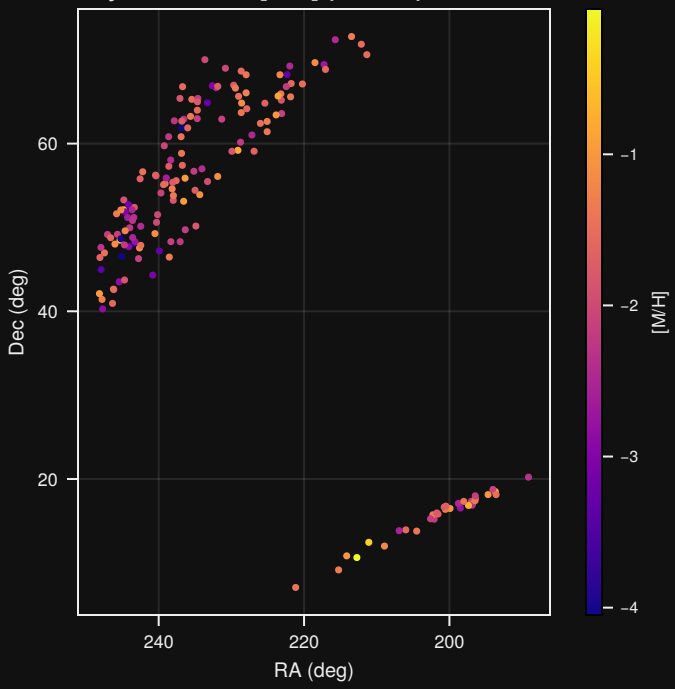
Star types from logg (n = 169)



Kiel diagram (n = 169)



Sky · GSP-Phot [M/H] (n = 169)

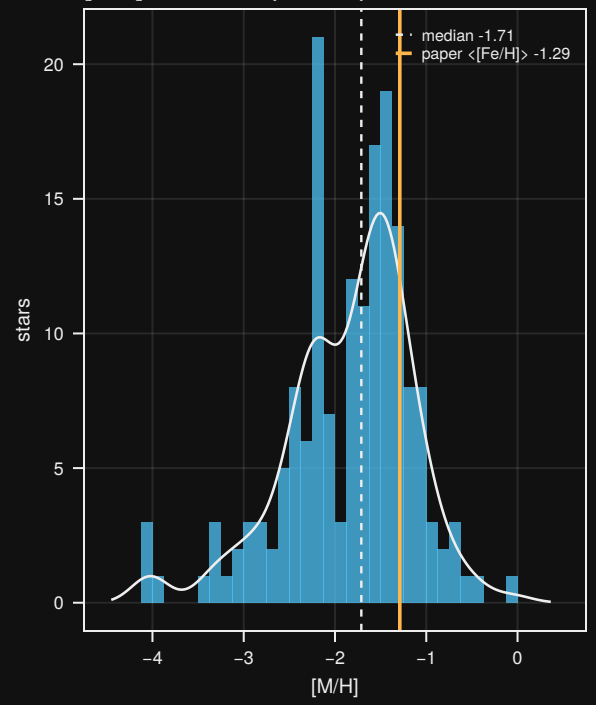


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≳ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

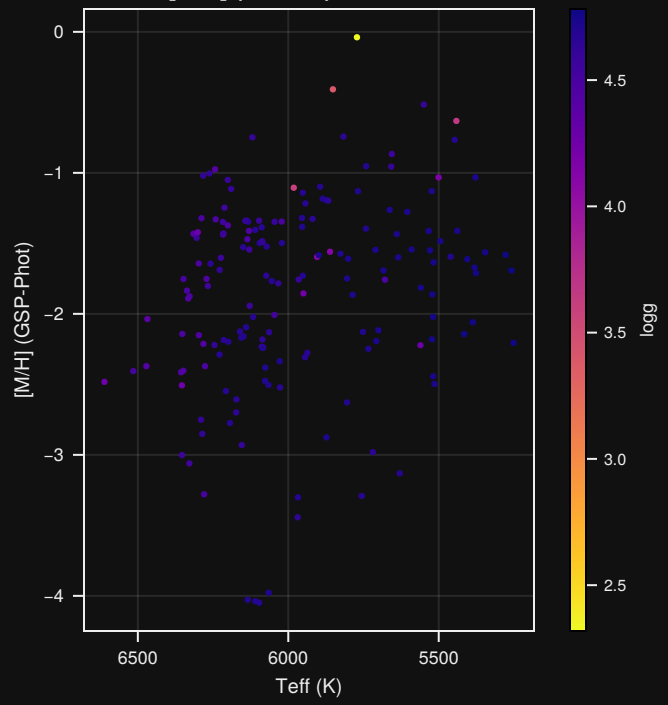
M5

Metallicity · age · astrometric quality

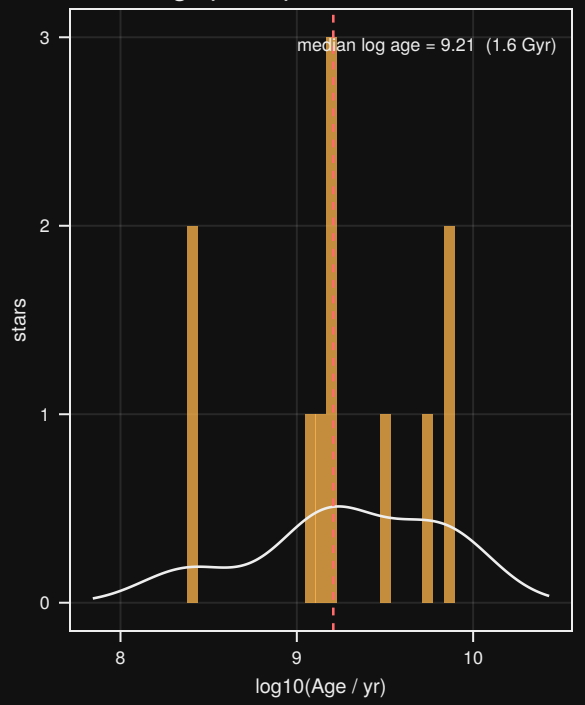
[M/H] GSP-Phot (n = 169)



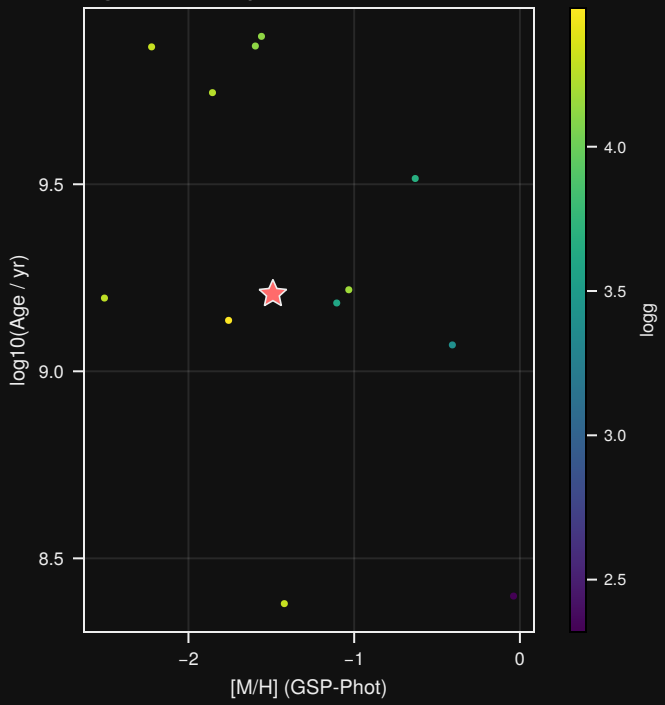
Teff vs [M/H] (n = 169)



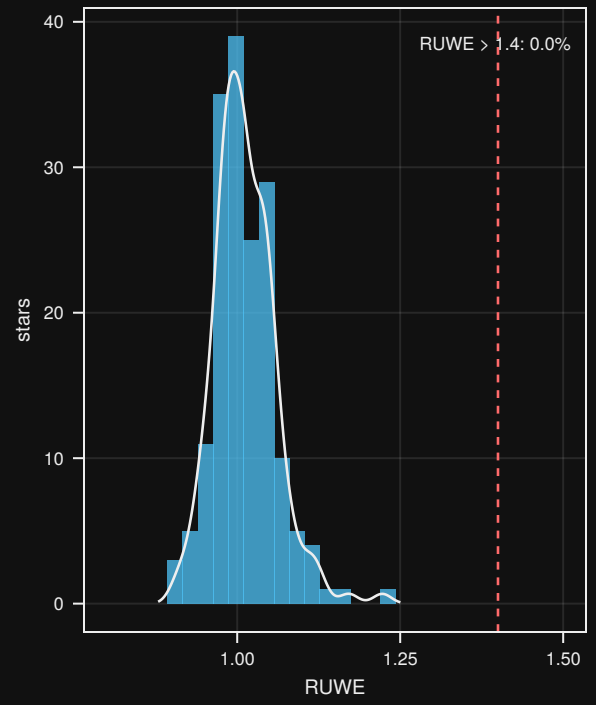
FLAME age (n = 12)



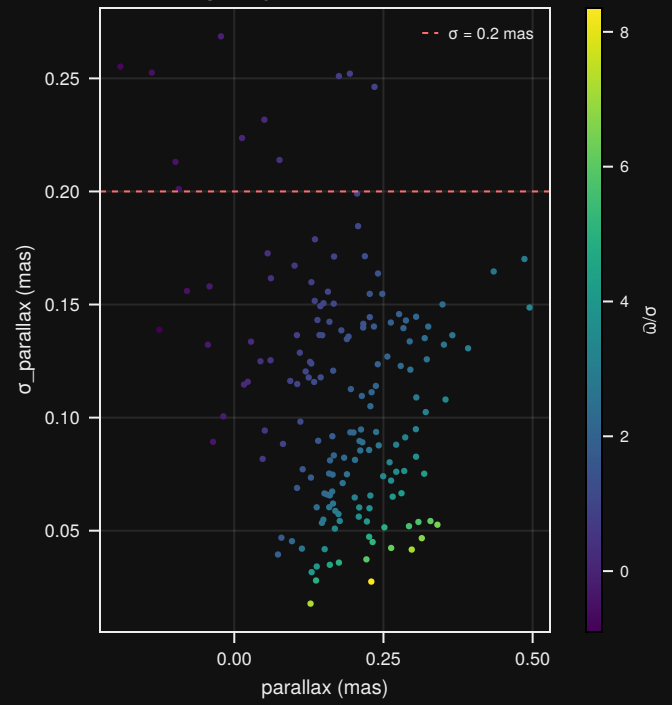
Age–metallicity (n = 12)



RUWE (n = 169)

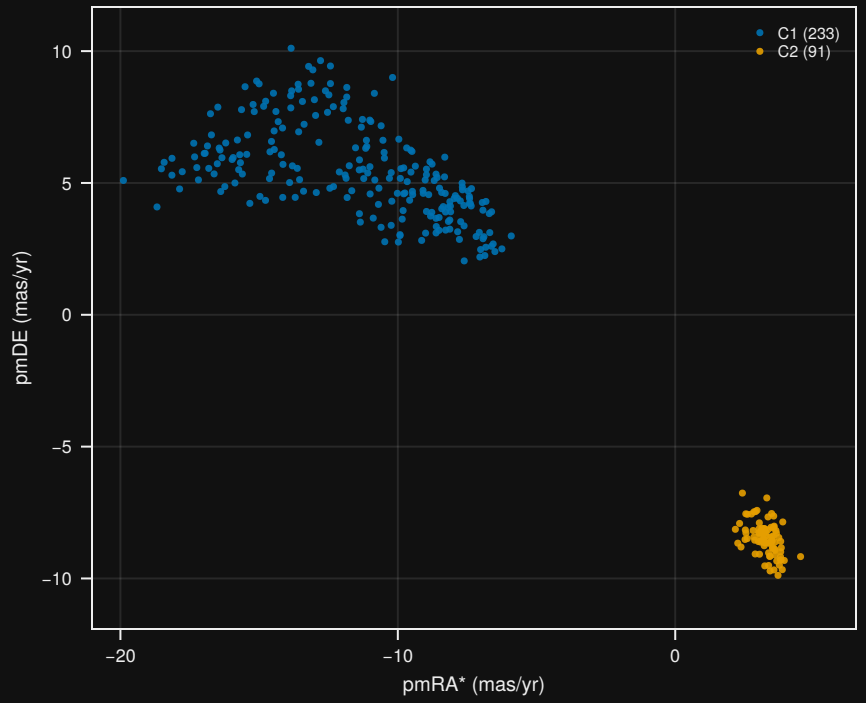


Parallax quality (n = 169)

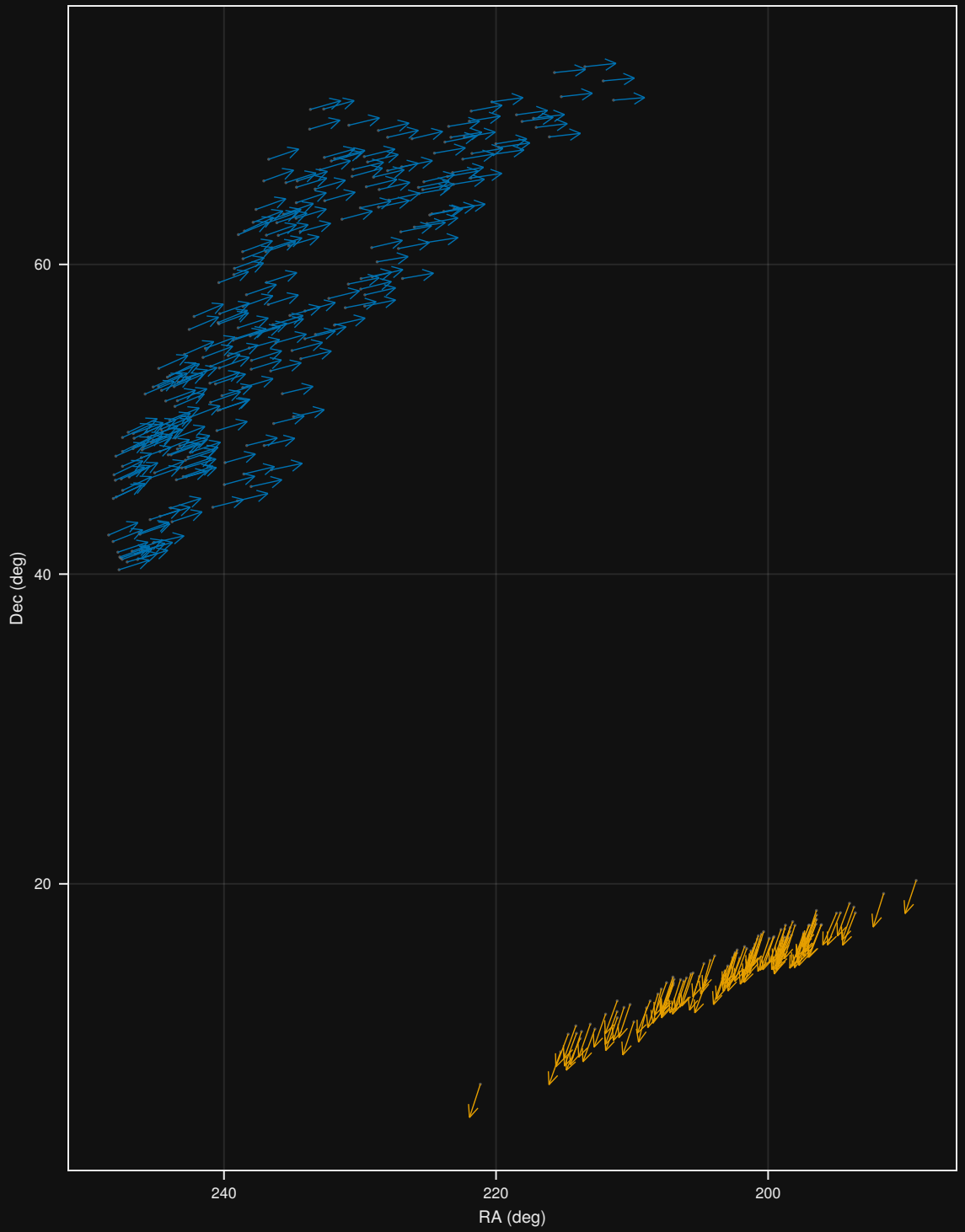


M5

PM clustering · HDBSCAN · 2 cluster(s) · noise 0%



Proper-motion directions (arrows) coloured by cluster



Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	0	—	—	—	—	0
C1	233	-11.01	5.21	4.01	-1.75	25
C2	91	3.27	-8.49	0.40	-1.43	11

M5

Per-cluster proper-motion vector maps

